

CO2 ASSESSMENT TOOL 1

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system.

```
# Verify that Nmap is installed
nmap --version

# Basic scan of the target
sudo nmap 192.168.1.10

# Scan the ports mentioned in the question and detect service versions
sudo nmap -sV -p 22,80,3306 192.168.1.10

# Service/version detection plus OS detection
sudo nmap -sV -O -p 22,80,3306 192.168.1.10

# Detailed authorized assessment scan
sudo nmap -sC -sV -O -p 22,80,3306 192.168.1.10

# Save the output for the report
sudo nmap -sC -sV -O -p 22,80,3306 -oN nmap_report.txt 192.168.1.10

# View the saved report
cat nmap_report.txt
```

2phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. Using the Go Phish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks

```
# Check whether GoPhish is present in the current directory
ls -lah

# If an authorized GoPhish package has already been downloaded/extracted,
# enter its directory (change the path to your lab installation)
cd ~/gophish

# Make the GoPhish binary executable if required
chmod +x gophish
```

```
# Start the local GoPhish lab console
sudo ./gophish

# In another terminal, confirm which local ports are listening
sudo ss -tulpn | grep -E ':(3333|80|443)\b'

# Optional: inspect the local GoPhish log generated by your lab
tail -f gophish.log
```

3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure.

```
# Replace example.com with the authorized lab/test domain

# Find the authoritative name servers
dig NS example.com

# Display common DNS records
dig A example.com
dig AAAA example.com
dig MX example.com
dig TXT example.com

# Show the SOA record
dig SOA example.com

# Test AXFR against a specific authoritative DNS server
# Replace ns1.example.com with a name server returned by the NS query
dig AXFR example.com @ns1.example.com

# Save the authorized test result
dig AXFR example.com @ns1.example.com > dns_axfr_report.txt

# View the saved report
cat dns_axfr_report.txt
```